

Weak Convergence of TEPROG Without Global Strict Convexity

Candidate full solution for the FA/Banach discovery run

22 July 2026

Source and target

The source is Reem, Reich, and De Pierro, *A Telescoping Bregmanian Proximal Gradient Method Without the Global Lipschitz Continuity Assumption*, arXiv:1804.10273v4 [1]. Remark 6.4, spanning pages 11–12, asks whether the weak convergence result continues to hold when the Bregman generator (b) is no longer strictly convex. The source PDF is included as `source_paper.pdf`. The two following full-width crops show the complete remark across the page break.

is uniformly convex on $\text{dom}(b)$, then the above-mentioned sequence $(x_k)_{k=1}^\infty$ converges weakly to a solution of (8).

Remark 6.4. It is possible to weaken a bit some of the assumptions needed for the definition of TEPROG and for the non-asymptotic convergence. Indeed, if b is semi-Bregman with the exception of being strictly convex on $\text{dom}(b)$, then $p_{L_k, \mu_k, S_k}(x_{k-1})$ will be a nonempty subset of S_k (this is a consequence of Lemma 8.3) and hence (by Assumption 5.4) of $S_k \cap U$, and x_k from either Algorithm 5.5 or Algorithm 5.6 can be taken to be any point in $p_{L_k, \mu_k, S_k}(x_{k-1})$. The proof of the non-asymptotic convergence in Theorem 6.1 remains as it is. However, it

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is an open problem whether the weak convergence result holds when b is no longer strictly convex, since the derivation of this convergence result is based on Lemma 8.1, which is based on the assumption that b is strictly convex.

The core question is whether “the weak convergence result holds when b is no longer strictly convex.” The natural literal reading retains the other hypotheses of Section 5 and Theorem 6.1, in particular Assumption 5.2: the nested telescopic sets S_k cover C , and b is strongly convex on each S_k , with parameter $\mu_k > 0$.

Status and novelty check

This packet gives a candidate full affirmative answer under the literal retained hypotheses. It also gives a finite-dimensional counterexample to the broader interpretation in which local strong convexity on the telescopic sets is removed together with global strict convexity.

A bounded search on 22 July 2026 used the exact paper title, arXiv id, the open-question phrase, `TEPROG weak convergence strict convexity`, and close variants. The 13 works in the OpenAlex citation list for DOI 10.1007/s10957-019-01509-8 were screened by title and available abstract or metadata. No later paper claiming this exact answer was found. This is negative search evidence, not a comprehensive novelty guarantee, so novelty confidence is moderate.

Framework

Let X be a reflexive Banach space. Let $b : X \rightarrow (-\infty, \infty]$ be proper, lower semicontinuous, and convex, and Gateaux differentiable on $U = \text{int}(\text{dom } b) \neq \emptyset$. We do *not* assume that b is strictly convex on all of $\text{dom } b$. For $x \in \text{dom } b$ and $y \in U$, set

$$B(x, y) = b(x) - b(y) - \langle b'(y), x - y \rangle.$$

Convexity implies $B(x, y) \geq 0$.

The remaining notation and assumptions are those of Section 5 of the source. In particular, C is closed and convex, $\text{MIN}(F) \neq \emptyset$ is contained in U , and

$$S_1 \subseteq S_2 \subseteq \cdots \subseteq C, \quad \bigcup_{k \geq 1} S_k = C,$$

where b is strongly convex on S_k with parameter $\mu_k > 0$. The step-size condition (20), the limiting difference property, and bounded first-type Bregman level sets are assumed exactly as in the weak-convergence part of Theorem 6.1.

Proof intuition

The global strict-convexity axiom is used in the source proof only to show that two weak cluster points cannot differ. But cluster points are not arbitrary points of $\text{dom } b$: the preceding part of the proof puts them in $\text{MIN}(F) \subset C$. Since the nested S_k 's cover C , any two cluster points lie together in one S_N . Strong convexity on that single set gives all the separation needed.

More explicitly, the source proof already gives a finite limit $\ell_q = \lim_k B(q, x_k)$ for every weak cluster point q . For two cluster points q_1, q_2 , the limiting difference property computes the two directed divergences as opposite real numbers. Both are nonnegative, so both are zero. Local strong convexity on a common S_N then turns zero divergence into $q_1 = q_2$. This is the whole missing step.

A localized cluster-point lemma

Proposition. *Let (x_k) be a bounded sequence in U . Suppose every weak cluster point belongs to a set $M \subset U$, the limit $\ell_q = \lim_{k \rightarrow \infty} B(q, x_k)$ exists and is finite for every weak cluster point q , and B has the limiting difference property. If for every two points $q_1, q_2 \in M$ there is a convex set $S \subset \text{dom } b$ containing them on which b is strictly convex, then (x_k) converges weakly.*

Proof. Let q_1 and q_2 be weak cluster points, obtained along subsequences N_1 and N_2 , respectively.

The limiting difference property gives

$$\begin{aligned} B(q_2, q_1) &= \lim_{\substack{k \rightarrow \infty \\ k \in N_1}} (B(q_2, x_k) - B(q_1, x_k)) = \ell_{q_2} - \ell_{q_1}, \\ B(q_1, q_2) &= \lim_{\substack{k \rightarrow \infty \\ k \in N_2}} (B(q_1, x_k) - B(q_2, x_k)) = \ell_{q_1} - \ell_{q_2}. \end{aligned}$$

Thus $B(q_2, q_1) = -B(q_1, q_2)$. Both terms are nonnegative by convexity, so both vanish. Strict convexity on a convex set containing q_1, q_2 , together with differentiability at the second argument, implies that zero Bregman divergence is possible only when $q_1 = q_2$. Hence all weak cluster points coincide.

It remains to upgrade this uniqueness to convergence. If the sequence did not converge weakly to its unique cluster point q , some weak neighborhood of q would omit an entire subsequence. That bounded subsequence has a weakly convergent further subsequence by reflexivity, whose limit must again be q , a contradiction. Therefore $x_k \rightharpoonup q$. $\square$

Main theorem: affirmative answer under the stated hypotheses

Theorem. *Retain the framework and assumptions of the weak-convergence part of Theorem 6.1 of [1], including Assumptions 5.1–5.4 and, in particular, strong convexity of b on every telescopic set S_k . Replace only the global strict-convexity requirement on $\text{dom } b$ by convexity. Then every TEPROG sequence generated by Algorithm 5.5 or 5.6 converges weakly to a point of $\text{MIN}(F)$. The same conclusion holds under the “in particular” sufficient conditions in Theorem 6.1.*

Proof. First note that the proximal subproblem on S_k is still uniquely solvable: strong convexity of b on S_k implies strict convexity there, so the objective defining p_{L_k, μ_k, S_k} is strictly convex. Thus deleting global strict convexity creates no selection ambiguity under Assumption 5.2.

All estimates in the proof of Theorem 6.1 through the construction of weak cluster points remain unchanged. They use convexity of b , strong convexity on the active sets S_k , and the other retained assumptions, but not strict convexity outside C . Concretely, those estimates establish:

- (i) (x_k) is bounded;
- (ii) every weak cluster point q belongs to $\text{MIN}(F) \subset C \cap U$;
- (iii) for every such q , the finite limit $\ell_q = \lim_k B(q, x_k)$ exists.

Apply the preceding proposition with $M = \text{MIN}(F)$. It remains only to verify its local strictness condition. Given $q_1, q_2 \in M \subset C$, choose indices i, j with $q_1 \in S_i$ and $q_2 \in S_j$. For $N = \max\{i, j\}$, nestedness gives $q_1, q_2 \in S_N$, and b is strongly convex on S_N .

For completeness, strong convexity yields the quantitative separation directly. For $0 < t < 1$,

$$b(q_1 + t(q_2 - q_1)) \leq (1 - t)b(q_1) + tb(q_2) - \frac{\mu_N}{2}t(1 - t)\|q_2 - q_1\|^2.$$

Subtract $b(q_1)$, divide by t , and let $t \downarrow 0$. Since $q_1 \in U$, Gateaux differentiability gives

$$B(q_2, q_1) \geq \frac{\mu_N}{2}\|q_2 - q_1\|^2.$$

The cluster-point calculation in the proposition gives $B(q_2, q_1) = 0$, hence $q_1 = q_2$. Therefore the whole sequence converges weakly, and its limit belongs to $\text{MIN}(F)$. $\square$

Remark. *The proof shows a sharper replacement for global strict convexity: it is enough that the Bregman divergence separate pairs of weak cluster points. Strong convexity on a nested cover of C is more than sufficient.*

Broader interpretation: failure without local separation

The wording of Remark 6.4 mentions choosing an arbitrary point from a possibly set-valued proximal map. Under Assumption 5.2 the map is actually singleton, as observed above. If the intended question also removes Assumption 5.2, the answer changes.

Theorem. *If local strong convexity (or another separation condition on cluster points) is deleted together with global strict convexity, arbitrary proximal selections need not converge weakly, even in dimension two with bounded U and a zero objective.*

Proof. Let $X = \mathbb{R}^2$ with its Euclidean norm and define

$$b(s, t) = \begin{cases} \frac{1}{2}s^2, & (s, t) \in [-2, 2]^2, \\ +\infty, & \text{otherwise.} \end{cases}$$

Then b is proper, lower semicontinuous, and convex, with bounded zone $U = (-2, 2)^2$, and $b'(s, t) = (s, 0)$ on U . Its derivative is weak-to-weak-star sequentially continuous, and

$$B((a, c), (s, t)) = \frac{1}{2}(a - s)^2.$$

Take $C = S_k = [-1, 1]^2$ for every k , and $f = g = 0$. Then $\text{MIN}(F) = C \subset U$, the gradient of f is Lipschitz, every proximal subproblem has a nonempty solution set contained in U , and the boundedness alternative for the weak-convergence hypotheses is automatic because U is bounded. For arbitrary positive L, μ , the proximal subproblem at $(s, t) \in C$ minimizes

$$\frac{L}{2\mu}(a - s)^2 \quad \text{over } (a, c) \in C.$$

Consequently its solution set is

$$\{(s, c) : -1 \leq c \leq 1\}.$$

Starting with $x_1 = (0, 1)$, choose successively $x_k = (0, (-1)^{k+1})$. Every x_k is an exact allowed proximal choice from the preceding point, but the sequence alternates between two points. In finite dimension weak and norm convergence coincide, so it has no weak limit.

This example intentionally violates Assumption 5.2: b is flat in the second coordinate and is not strongly convex on C . It therefore does not contradict the affirmative theorem; it identifies precisely why the retained local hypothesis matters. $\square$

Verification notes and limitations

- The proof is analytic; no numerical experiment is used as evidence.
- The source proof of Lemma 8.1 was checked line by line. Its global strict-convexity invocation occurs only after the limiting-difference calculation; the local estimate in the main theorem replaces exactly that step.
- The evidence crops are reproducible from the packet directory with

```
conda run -n sandbox python
code/make_open_problem_crops.py
```

- Human review should focus on the interpretation of Remark 6.4 and on the fact that Assumption 5.2 is formally retained. Both interpretations are mathematically covered here.
- The novelty search was bounded and cannot exclude an unindexed answer.

References

- [1] D. Reem, S. Reich, and A. De Pierro, *A Telescoping Bregmanian Proximal Gradient Method Without the Global Lipschitz Continuity Assumption*, Journal of Optimization Theory and Applications 182 (2019), 851–884; arXiv:1804.10273v4.